



BIO411/BIO629 1st Mid-semester Exam (2025) IISER Mohali

Date: 02/09/2025

Duration: 60 min

Maximum marks: 24 ~~25~~

Instructions:

- All questions are compulsory.
- Questions in Section A carry 1 marks.
- Answer questions briefly in section B. Include equations, wherever applicable.

Section A

- In DNA sequencing output, Q-score of 30 indicates *quality measure of an individual base call in a sequencing read (probability of incorrect base call), which is 0.001 error probability and accuracy is 99.9%.*
- MeSH terms in MEDLINE is defined as *Medical Subject Heading controlled vocabulary thesaurus used for indexing, cataloging, and searching life science and biomedical information.*
- HiFi read in the context of PacBio sequencing refers to *High fidelity reads is a type of sequencing read generated by circular re-sequencing provides long length and high accuracy.*
- The CGR acronym stands for *Chaos Game Representation*. In CGR representation of a DNA sequence with repeating 'C' nucleotide *will be along diagonal in the C nucleotide quadrant.*
- The Information content (IC) for the sequence "GGGGGGGG" is *2 bits* and "CGTAATCG" is *0 bits.*
- In skin microbiome sequencing, a scaffold is found to have (GC)% of 23%. Is the scaffold sequence likely from human genome? (True/False). *You can make assumption. False (human genome GC% is 41%)*
- Huffman encoded string 01010010 represents DNA sequence *GAAGA*. Single nucleotide composition is A:10, C:8, G:12, T:2
- For the given sequence **X**, the %GC content is *55%* Relative dinucleotide frequency or abundance of TA is $(2/19)/((4/20)*(5/20)) \sim 2.1$ (assume single strand in counting; S. Karlin approach).

Sequence **X** = **GCGCATGCGGCTATAT**

Section B

- Describe briefly any one Next Generation Sequencing method. **(2 marks)**
- In a search for a new TBFS in the genome of *Streptomyces coelicolor*, an organism with an exceptionally high GC-content (~72%) using Gibbs sampler resulted in top scoring motif as GCGGCG. Which will be preferred background model to evaluate statistical significance of the motif? Provide suitable reason for the same. **(2 marks)**

As *S. coelicolor* has high GC content, using intergenic composition model may over estimate significance of the motif. One can use background model:

- 1st or higher order Markov chain background model can be used.

11. Burrows-Wheeler Transform of the sequence ATAGGC. (2 marks)

CT\$GGAA

12. Write various ways of representing DNA sequence motif. (3 marks)

Position Frequency Matrix

Position weight matrix

Sequence logos

Consensus sequence/Degenerate /Regular expression

13. Mr. X has identified 30 co-expressed yeast genes under oxidative stress. Suggest any one *in silico* approach (method) to identify putative transcription factor binding sites. Provide various algorithmic steps of method and limitations of the approach. (3 marks)

Gibbs sampling

EM algorithm

Oligo-analysis

14. Write probability of the given sequence (**S**) for questions to a-c under following assumption: (5 marks)

Sequence **S** = GTAGTTCGTA

a. Equal probable or null nucleotide model. (1)

$$P(S) = (1/4)^{10}$$

b. Motif represented as PSSM (given below) starts at position 2. (1)

$$P(S) = p_{G,0} \times p_{T,1} \times p_{A,2} \times p_{G,3} \times p_{T,4} \times p_{T,0} \times p_{C,0} \times p_{G,0} \times p_{T,0} \times p_{A,0}$$

$$P(S) = 0.1 \times 0.1 \times 0.1 \times 0.5 \times 0.6 \times 0.4 \times 0.3 \times 0.1 \times 0.4 \times 0.2$$

c. ~~The background sequence of DNA is modeled as 1st order markov chain and motif (4 character) starts at 4th position in the sequence **S**. (Transition probabilities are given below):~~

d. Matching position of motif in the sequence (**S**) with highest likelihood considering given PSSM motif model. (2)

Scanning Motif on the sequence at position 1 and compute P(S):

Pos1: GTAGTTCGTA

Pos2: GTAGTTCGTA

Pos3: GTAGTCGTA (highest score)

Pos4: GTAGTTCGTA

Pos5: GTAGTTCGTA

Pos6: GTAGTTCGTA

Pos7: GTAGTTCGTA

PSSM of a motif

	0	1	2	3	4
A	0.20	0.7	0.1	0.1	0.2
C	0.30	0.1	0.1	0.1	0.1
G	0.10	0.1	0.6	0.5	0.1
T	0.40	0.1	0.2	0.3	0.6

Transition probability matrix

	B	M
B	0.8	0.2
M	0.3	0.7
0 (Start)	0.9	0.1

B-background

M-Motif